

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0628

Speed 138 kV
October 25, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0628. The Interconnection Customer proposes a 284.2 MW natural gas combined cycle facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on June 11, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 272 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$45,901,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SANTA MONICA-HAMBRICK 138 kV
Contingency: SILVER BAY POWER-SPEED SUBSTATION 138 kV Line
Loading: 225.4% (Rating: 218 MVA)
DFAX: 100.0%
Pre-project loading: 95.3%
2. Facility: NAROD-BROCKPORT 138 kV
Contingency: SPEED-SILVER BAY POWER 138 kV
Loading: 214.2% (Rating: 221 MVA)
DFAX: 100.0%
Pre-project loading: 85.8%
3. Facility: 56TH ST-SANTA MONICA 138 kV
Contingency: SILVER BAY POWER-SPEED 138 kV
Loading: 207.8% (Rating: 228 MVA)
DFAX: 100.0%
Pre-project loading: 82.9%
4. Facility: FERNWOOD-CHESTNUT 138 kV
Contingency: CONSOLIDATED COAL-CHESTNUT 138 kV
Loading: 203.6% (Rating: 202 MVA)
DFAX: 7.4%
Pre-project loading: 193.1%
5. Facility: NAROD-BARNHAM 138 kV
Contingency: SPEED-SILVER BAY POWER 138 kV
Loading: 197.9% (Rating: 256 MVA)
DFAX: 100.0%
Pre-project loading: 87.0%

6. Facility: BARNHAM-GAGA 138 kV
Contingency: SPEED-SILVER BAY POWER 138 kV L
Loading: 189.1% (Rating: 278 MVA)
DFAX: 100.0%
Pre-project loading: 87.0%
7. Facility: DAVISVILLE-MIRABEL 138 kV line
Contingency: WELLINGTON-DAVISVILLE 138 kV
Loading: 186.2% (Rating: 217 MVA)
DFAX: 5.9%
Pre-project loading: 178.5%
8. Facility: MIRABEL-WEC 138 kV
Contingency: WELLINGTON-DAVISVILLE 138 kV Line
Loading: 185.2% (Rating: 222 MVA)
DFAX: 5.9%
Pre-project loading: 177.6%
9. Facility: SILVER BAY POWER-SPEED 138 kV
Contingency: GAGA-SPEED 138 kV
Loading: 181.9% (Rating: 299 MVA)
DFAX: 100.0%
Pre-project loading: 87.0%
10. Facility: SPEED-GAGA 138 kV L
Contingency: SILVER BAY POWER-SPEED 138 kV
Loading: 181.9% (Rating: 299 MVA)
DFAX: 100.0%
Pre-project loading: 87.0%
11. Facility: 56TH STREET-OATES REIT 138 kV line
Contingency: SILVER BAY POWER-SPEED 138 kV
Loading: 178.3% (Rating: 247 MVA)
DFAX: 100.0%
Pre-project loading: 63.1%
12. Facility: SAND SPRINGS-C-33 DOE 138 kV
Contingency: DAVISVILLE-N APPLETON SUBSTATION 138 kV Line
Loading: 158.4% (Rating: 390 MVA)
DFAX: 5.4%
Pre-project loading: 154.4%
13. Facility: C-33 DOE-PLATTE CITY 138 kV
Contingency: DAVISVILLE-NORTH APPLETON 138 kV
Loading: 156.4% (Rating: 400 MVA)
DFAX: 5.4%
Pre-project loading: 152.6%
14. Facility: PLATTE CITY-ALBION RD 138 kV
Contingency: DAVISVILLE-NORTH APPLETON 138 kV
Loading: 154.2% (Rating: 413 MVA)
DFAX: 5.4%
Pre-project loading: 150.6%
15. Facility: ST JOSPH SUBSTATION-CHESTNUT 138 kV
Contingency: BARNABIE CABLE STA-CASINO 138 kV
Loading: 154.2% (Rating: 396 MVA)
DFAX: 7.2%
Pre-project loading: 149.0%
16. Facility: LANHM-RANCHO CARMEL 138 kV
Contingency: LIVERMORE-LEE COUNTY SOLID WASTE ENERGY 345 kV

- Loading: 151.9% (Rating: 238 MVA)
DFAX: 5.9%
Pre-project loading: 144.9%
17. Facility: ST JOSEPH-LESLIE SUBSTATION 138 kV
Contingency: BARNABIE CABLE STA-CASINO 138 kV
Loading: 151.4% (Rating: 413 MVA)
DFAX: 7.2%
Pre-project loading: 146.5%
18. Facility: N APPLETON-LESLIE SUBSTATION 138 kV
Contingency: CASINO-BARNABIE CABLE STA 138 kV
Loading: 150.3% (Rating: 420 MVA)
DFAX: 7.2%
Pre-project loading: 145.5%
19. Facility: CHESTNUT-CONSOLIDATED COAL 138 kV
Contingency: CHESTNUT-FERNWOOD 138 kV
Loading: 150.0% (Rating: 242 MVA)
DFAX: 6.3%
Pre-project loading: 142.6%
20. Facility: NORTH APPLETON-DAVISVILLE 138 kV Line
Contingency: BARNABIE CABLE STA-CASINO 138 kV
Loading: 148.9% (Rating: 430 MVA)
DFAX: 7.2%
Pre-project loading: 144.1%
21. Facility: CPU SILICON-LANHAM 138 kV L
Contingency: LIERMORE-LEE COUNTY SOLID WASTE ENERGY 345 kV
Loading: 142.1% (Rating: 175 MVA)
DFAX: 6.2%
Pre-project loading: 132.1%
22. Facility: MAINWARING-SNOOMISH 138 kV
Contingency: SAND SPRINGS-OJO SUBSTATION 138 kV L
Loading: 141.2% (Rating: 259 MVA)
DFAX: 7.0%
Pre-project loading: 133.5%
23. Facility: HOLDEN-EMBUDO 138 kV
Contingency: LIVERMORE-LEE COUNTY SOLID WASTE ENERGY 345 kV Line
Loading: 136.7% (Rating: 236 MVA)
DFAX: 6.2%
Pre-project loading: 129.2%
24. Facility: SILVER BAY POWER-HOTEL ROD 138 kV
Contingency: LIVERMORE-LEE COUNTY SOLID WASTE ENERGY 345 kV
Loading: 135.4% (Rating: 224 MVA)
DFAX: 22.5%
Pre-project loading: 106.9%
25. Facility: CPU SILICON-KINGS ROW 138 kV Line
Contingency: LIVERMORE-LEE COUNTY SOLID WASTE ENERGY 345 kV Line
Loading: 131.9% (Rating: 202 MVA)
DFAX: 6.2%
Pre-project loading: 123.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SANTA MONICA-HAMBRICK 138 kV	Rebuild line with higher-capacity conductor	\$3,267,000	50% (\$1,649,000)
NAROD SUBSTATION-BROCKPORT 138 kV	Rebuild line with higher-capacity conductor	\$4,345,000	50% (\$2,193,000)
56TH ST-SANTA MONICA 138 kV	Rebuild line with higher-capacity conductor	\$3,107,000	50% (\$1,569,000)
FERNWOOD-CHESTNUT 138 kV L	Rebuild line with higher-capacity conductor	\$9,371,000	8% (\$715,000)
NAROD-BARNHAM 138 kV	Rebuild line with higher-capacity conductor	\$17,150,000	50% (\$8,657,000)
GAGA-BARNHAM 138 kV	Rebuild line with higher-capacity conductor	\$2,585,000	50% (\$1,305,000)
MIRABL-DAVISVILLE 138 kV	Rebuild line with higher-capacity conductor	\$27,615,000	7% (\$2,050,000)
MIRABEL-WEC 138 kV	Rebuild line with higher-capacity conductor	\$7,898,000	7% (\$586,000)
SILVER BAY POWER-SPEED 138 kV	Rebuild line with higher-capacity conductor	\$17,778,000	50% (\$8,974,000)
GAGA-SPEED 138 kV	Rebuild line with higher-capacity conductor	\$2,662,000	50% (\$1,344,000)
56TH STREET-OATES REIT 138 kV line	Rebuild line with higher-capacity conductor	\$2,961,000	50% (\$1,495,000)
SAND SPRINGS-C-33 DOE 138 kV	Rebuild line with higher-capacity conductor	\$7,267,000	5% (\$339,000)
C-33 DOE-PLATTE CITY 138 kV	Rebuild line with higher-capacity conductor	\$5,168,000	5% (\$241,000)
ALBION RD SUBSTATION-PLATTE CITY 138 kV	Rebuild line with higher-capacity conductor	\$17,117,000	5% (\$799,000)
ST JOSEPH-CHESTNUT 138 kV Line	Rebuild line with higher-capacity conductor	\$6,366,000	6% (\$399,000)
LANHAM-RANCHO CARMEL 138 kV L	Rebuild line with higher-capacity conductor	\$6,727,000	7% (\$492,000)
ST JOSEPH-LESLIE SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$8,403,000	6% (\$527,000)
NORTH APPLETON-LESLIE 138 kV	Rebuild line with higher-capacity conductor	\$4,998,000	6% (\$313,000)
CHESTNUT-CONSOLIDATED COAL 138 kV	Rebuild line with higher-capacity conductor	\$39,658,000	8% (\$2,976,000)
NORTH APPLETON-DAVISVILLE 138 kV Line	Rebuild line with higher-capacity conductor	\$4,785,000	6% (\$300,000)
CPU SILICON-LANHAM 138 kV	Rebuild line with higher-capacity conductor	\$4,709,000	13% (\$597,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SNOHOMISH-MAINWARING 138 kV L	Rebuild line with higher-capacity conductor	\$5,338,000	10% (\$530,000)
EMBUDO-HOLDEN 138 kV	Rebuild line with higher-capacity conductor	\$6,104,000	13% (\$773,000)
SILVER BAY POWER-HOTEL ROD 138 kV	Rebuild line with higher-capacity conductor	\$16,324,000	30% (\$4,907,000)
CPU SILICON-KINGS ROW 138 kV Line	Rebuild line with higher-capacity conductor	\$17,132,000	13% (\$2,171,000)
	Total allocated to project		\$45,901,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.